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Filter housing with drop out chamber and methods

Abstract

A filter housing includes a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume;) a filter element in the interior volume; the filter element having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; a lid removably mounted on the container to cover the access opening; and a gas inlet and gas outlet in at least one of the container and lid; the gas inlet and gas outlet being co-planar. The gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.

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Background/Summary

(1) This application claims the benefit of U.S. Provisional Patent Application No. 63/177,681, filed Apr. 21, 2021, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) This disclosure concerns a filter housing in a filter assembly for use in capturing hazardous particulate and gaseous/vaporized matter. In particular, this disclosure concerns a filter housing with a drop out chamber and methods.

BACKGROUND

(2) In many industries such as electronics fabrication, laser marking, laser cutting, engraving, and pharmaceuticals, extraction systems are used to capture hazardous particulate and gaseous or vaporized matter generated by industrial processes.

(3) Such systems can include particulate filters and/or a gas filter housed in a single unit, together with either a blower or a pump. The blower or pump will draw contaminated air into the unit and through the filters.

(4) While a variety of filter assemblies have been used, improvements are desirable. For example, the internal filter element can be expensive, and ways to extend the life of the filter element are desirable.

SUMMARY

(5) A filter housing and methods are provided to improve the prior art.

(6) In one aspect, a filter housing is provided comprising: (a) a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume; (b) a filter element in the interior volume; the filter element having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; (c) a lid removably mounted on the container to cover the access opening; and (d) a gas inlet and gas outlet in at least one of the container and lid; the gas inlet and gas outlet being coplanar. The gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.

(7) Preferably, the lid includes the gas inlet and the gas outlet.

(8) In example embodiments, the gas outlet includes an outlet tube extending into the interior volume of the container; and the filter element has an outlet opening receiving the outlet tube.

(9) In many example implementations, the filter element includes: (a) a media pack having the inlet face and the outlet face; and (b) a casing holding the media pack; wherein the casing includes an outlet wall spaced from the outlet face; the outlet wall defining the outlet opening of the filter element.

(10) In some examples, the casing has a surrounding wall extending between the outlet wall and the inlet face of the media pack.

(11) In preferred implementations, the surrounding wall of the casing provides a baffle for inlet flow from the gas inlet to the inlet face of the media pack.

(12) In one or more example embodiments, the media pack comprises pleated media.

(13) Preferably, the gas inlet and the filter element are arranged such that inlet gas turns between 160°-200° to flow from the gas inlet to the inlet face of the filter element.

(14) In one or more example embodiments, the container includes a ledge protruding into the interior volume constructed and arranged to hold the filter element.

(15) In example implementations, the container includes a filtering compartment and a debris collector disposed below the filtering compartment; the filtering compartment and debris collector separated by a sloping wall having an aperture; the aperture covered by an openable and closeable fitment.

(16) In preferred embodiments, a seal arrangement between the compartment and the lid, and a clamp arrangement providing a releasable clamping force holding the compartment and lid together in sealing engagement.

(17) In another aspect, a method of filtering a gas is provided. The method includes: (a) providing a container, a filter element in an interior volume of the container; and a lid mounted on the container; and (b) allowing gas to flow through a gas inlet, turn at least 90°, flow into the filter element, and then exit through a gas outlet. The inlet and gas outlet are co-planar.

(18) In example methods, the gas turns between 160°-200° to flow into the filter element.

(19) Preferably, the filter element includes a media pack having the inlet face and the outlet face; and a casing holding the media pack; wherein the casing has an outlet wall and a surrounding wall, the surrounding wall extending between the outlet wall and the inlet face of the media pack; and the step of allowing gas to flow includes the surrounding wall being a baffle for inlet flow from the gas inlet to the inlet face of the media pack.

(20) In example methods, the container includes a filtering compartment and a debris collector disposed below the filtering compartment; and while the gas flows from the gas inlet to the filter element, debris drops by gravity to a bottom of the filtering compartment.

(21) A variety of examples of desirable product features or methods are set forth in the description that follows, and in part, will be apparent from the description, or may be learned by practicing various aspects of this disclosure. The aspects of this disclosure may relate to individual features as well as combinations of features. It is to be understood that both the foregoing general description and the following detailed description are explanatory only, and are not restrictive of the claimed invention

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a filter assembly including a filter housing, constructed in accordance with principles of this disclosure;
- (2) FIG. 2 is an exploded, perspective view of the filter assembly of FIG. 1, in which a filter element can be seen exploded out of the filter housing;
- (3) FIG. 3 is a perspective, cross-sectional view showing gas flow through the filter assembly of FIG. 1;
- (4) FIG. 4 is a top perspective view of the filter element shown in FIG. 2;
- (5) FIG. 5 is a bottom perspective view of the filter element shown in FIG. 4;
- (6) FIG. 6 is a cross-sectional view of the filter assembly of FIG. 1 and showing an enlarged view of a portion of the seal arrangement between a lid and a container of the filter housing;
- (7) FIG. 7 is another cross-sectional view of the filter assembly of FIG. 1 and showing an enlarged view of a seal arrangement between an outlet tube of the filter housing and an outlet opening in the filter element;
- (8) FIG. 8 is a top plan view of one of the filtered gaskets used in the seal arrangement between the lid and container depicted in FIG. 6;
- (9) FIG. 9 is a front view of the gasket of FIG. 8; and
- (10) FIG. 10 is a perspective view of the gasket of FIG. 8.

DETAILED DESCRIPTION

(11) FIGS. 1-3 depict an example embodiment of a filter assembly 20. The filter assembly 20 can be used in a variety of systems, including closed loop systems. In closed loop systems, gas, such as air, containing particulate and gaseous or vaporized matter is taken into the filter assembly 20, filtered to remove the particulate, and then the filtered gas is exhausted. This exhausted gas may again be used in the overall system, without being expelled to the ambient atmosphere. Examples

of these types of systems include electronics fabrication, laser marking, laser cutting, engraving, and pharmaceuticals. The filter assembly **20** disclosed herein can be helpful in providing for safety when filtering and transporting filtered high risk particulate in controlled atmospheres. This solution is provided by the use of a seal arrangement, which shares differential pressure load. The sharing of the differential pressure load lowers the risk of leakage across the seal arrangement in applications such as 3D printing of metal powders, which are known as pyrophoric. In addition, the filter assembly **20** disclosed herein can be helpful in extending the life of the internal filter element.

(12) In FIGS. **1-3**, the filter assembly **20** includes a filter housing **22**. The filter housing **22** includes, collectively, a container **24** and a lid **26**.

(13) The container **24** includes a surrounding wall **28** defining an open interior volume **30** (FIG. **2**). The surrounding wall **28** is depicted as generally box or rectangular shaped, but it could be other shapes.

(14) In the example shown, the container **28** has a front wall **32**, an opposite back wall **34**, a first side wall **36**, and an opposite second side wall **38**. The first side wall **36** and second side wall **38** extend between and join the front wall **32** and back wall **34**.

(15) A terminal end of each of the front wall **32**, back wall **34**, first side wall **36**, and second side wall **38** forms an end rim **40**. The end rim **40** has a generally horizontal support surface **41** defines an access opening **42**. The access opening **42** is in communication with the interior volume **30** and allows access into the interior volume **30**.

(16) Opposite of the access opening **42** is a bottom wall **44** (FIG. **3**). The bottom wall **44** is sloping, or sloped, from each of the walls **32**, **34**, **36**, **38** to form a generally shaped funnel portion **46**. The funnel portion **46** leads to an aperture **48** in the bottom wall **44**. In the example shown, the aperture **48** is generally centered along the bottom wall **44**. The aperture may be covered by an openable and closable fitment **50**. The fitment **50** can be opened to drain any collecting debris resting on the bottom wall **44**.

(17) The container **24** will typically be made from a hard, rigid material, such as metal.

(18) In reference again to FIGS. **1** and **2**, the lid **26** is depicted removably mounted on the end rim **40** of the container **24** to cover the access opening **42**. The lid **26** is shown to be completely removable from the housing **24**, although other embodiments, it could be pivotably mounted relative to the container **24**.

(19) The lid **26** includes a main section **52**, which covers the access opening **42**, and a surrounding skirt **54** extending generally perpendicular from the main section **52**. The skirt **54** extends and overlaps the portion of the container **24** adjacent the access opening **42**, when the lid **26** is in covering relation to the access opening **42**.

(20) A gas inlet **56** and gas outlet **58** are provided in at least one of the container **24** and the lid **26**. In preferred implementations, the apertures for the gas inlet **56** and gas outlet **58** are contained in a same plane, i.e. are co-planar. The gas inlet **56** is surrounded by a gas inlet tube **57**, and the gas outlet **58** is surrounded by a gas outlet tube **59**. The outlet tube **59** extends into the interior volume **30** of the container **24**.

(21) In the example embodiment shown, the lid **26** includes the gas inlet **56** and gas outlet **58** extending through the main section **52**. When the lid **26** is mounted on the container **24**, interior ends of the gas inlet **56** and gas outlet **58** extend or project into the interior volume **30** of the container **24**. Exterior ends of the gas inlet **56** and gas outlet **58** are constructed and arranged to be releasably clamped to tubes or hoses that are in communication with the system being filtered.

(22) The lid **26** can further include a handle **60**. The handle **60**, in this embodiment, is shown connected to the skirt **54** and is helpful in removing the lid **26** from the container **24**.

(23) In FIGS. **1-3**, the housing **22** further includes a clamp arrangement **90**. The clamp arrangement **90** provides a releasable clamping force between the container **24** and the lid **26**. While many alternatives are possible, in the example shown, the clamp arrangement **90** includes a plurality of clamp levers **92** attached to the container **24** adjacent the access opening **42**, which engage a

plurality of catches **94** (receivers) secured to the skirt **54** of the lid **26**. The levers **92** control u-shaped arms **96** (FIG. **6**) that can be selectively engaged or disengaged from the catches **94**. While many variations are possible, in the example shown, the clamp arrangement **90** shown are latch style toggle clamps.

(24) In FIGS. **1** and **2**, it can be seen how the filter assembly **20** is portable by being mounted on a wheeled cart **62**. The container **24** is supported or held by the cart **62** and may be moved to a desired location using the wheels **64** to roll the cart **62** to the desired location.

(25) In reference to again to FIG. **2**, the filter assembly **20** includes a filter element **70**. The filter element **70** is positioned or oriented within the interior volume **30** of the container **24** for filtering gas that is drawn into the interior volume **30**. An example filter element is shown in more detail in FIG. **3**.

(26) In FIG. **3**, the filter element **70** includes a media pack **72**. The media pack **72** can be made from a variety of filter media. This example, the media pack **72** is made from pleated media, such as pleated paper. Others types of media are usable including Z-media.

(27) Z-media can include fluted, such as corrugated or pleated, media secured to a facing sheet. Typically, the facing sheet is non-fluted, non-corrugated. The Z-filter media can form a set of longitudinal (axial) flutes or air flow channels on one side of the corrugated or fluted media, and another set of longitudinal (axial) flow channels on an opposite side of the fluted media. The term “axial” in connection with the definition of longitudinal flutes is meant to refer to a direction of flute extensions between opposite flow faces. Flutes of one set of flutes are inlet flutes and are left open at the inlet end side of the media and are sealed or otherwise folded closed at an outlet end side of the media. Analogously, the flutes of a second set of flutes are outlet flutes and are sealed or otherwise closed at the inlet end side of the media, and are left open at the outlet end of the media. In operation, the air passes into the inlet flow face of the media pack by passage into the open inlet flutes at an upstream face of the filter. The air cannot flow out of the closed ends of these inlet flutes, so it must pass through the filter media into the outlet flutes. The filtered air then passes outwardly from the outlet flutes and through the downstream flow face.

(28) The filter element **70** includes an inlet face **74** and an opposite outlet face **76**. In this embodiment, a casing **78** holds the media pack **72** around the sides between the inlet face **74** and outlet face **76**. Gas to be filtered flows into the inlet face **74**, through the pleated media, and exits the filter element by flowing out of the outlet face **76**. The pleated media removes particulate from the gas stream.

(29) The filter element **70** includes an outlet opening **80**. The casing **78** includes an outlet wall **82** spaced from the outlet face **76**. The outlet wall **82** defines the outlet opening **80**.

(30) The outlet opening **80** includes a surrounding inner rim **84** (FIG. **7**). The outlet opening **80** is sized to receive the outlet tube **59** from the lid **26**. In FIG. **7**, it can be seen how the outlet opening **80** of the filter element **70** includes a seal member **86**, for example a radial seal member **86**, along the inner rim **84** of the outlet opening **80**. The seal member **86** forms a seal by compression of the radial seal member **86** between and against the inner rim **84** of the outlet opening **80** and the outlet tube **59**.

(31) In reference again to FIGS. **4** and **5**, the casing **78** of the filter element **70** further includes a pair of handles **88**. The handles **88** extend or project above a plane containing the outlet wall **82** of the casing **78** and can be grasped in order to move the filter element **70** into or out of the interior volume **30** of the container **24**.

(32) In reference to FIG. **3**, inlet gas is shown at arrow **98**. As can be seen in FIG. **3**, the gas inlet **56** and the filter element **70** are arranged such that inlet gas **98** turns at least 90° to flow from the gas inlet **56** to the inlet face **74** of the filter element **70**. In preferred embodiments, the gas inlet **56** and the filter element **70** are arranged such that inlet gas **98** turns between 160°-200° (i.e., about 180°) to flow from the gas inlet **56** to the inlet face **74** of the filter element **70**.

(33) In preferred arrangements, an inlet cavity **99** (FIG. **3**) of the container **24** is of a greater cross

sectional area than that of the gas inlet tube **57**, and therefore, the flow velocity of the gas inlet air **98** drops. Additionally, the flow changes direction by 180 degree turn to enter the inlet flow face **74** of the filter element **70**. This change in direction further slows the flow velocity of the gas inlet air **98** and further provides opportunity for suspended heavy particulate to drop from the gas inlet air **98**. The particulate drops to the funnel portion **46** of the bottom wall **44** of the container **24**, and can be selectively removed by opening the fitment **50** of the aperture **48**.

(34) From a review of FIG. **3**, the casing **78** of the element **70** can function as a baffle for the inlet flow **98** from the gas inlet **56** to the inlet face **74** of the media pack **72**.

(35) Still in reference to FIG. **3**, in preferred implementations, the container **24** includes a ledge **25** protruding into the interior volume **30** constructed and arranged to hold the filter element **70**. Many variations are possible.

(36) The filter assembly **20** further includes a seal arrangement **100** (FIG. **6**). The seal arrangement **100** is between the lid **26** and the container **24** mounted to form a releasable seal therebetween. The seal arrangement **100** includes a container gasket **102**. The container gasket **102** is secured to the container **24** at a position surrounding the access opening **42**. In this example, the container gasket **102** is secured on an axial surface of the support surface **41** of the container **24**.

(37) The container gasket **102** includes a sealing portion **104** which is oriented in a direction toward the lid **26**. The container gasket **102** defines a first longitudinal axis **108** (FIG. **6**) passing through a center of the cross-sectional width of the gasket **102**. The longitudinal axis **108**, in this example, is generally parallel to the first side wall **36** of the surrounding wall **28** of the container **24**.

(38) The seal arrangement **100** further includes a lid gasket **110**. The lid gasket **110** has a sealing region **112** oriented in a direction toward the container **24**. The lid gasket **110** has a second longitudinal axis **114** (FIG. **6**) passing through a center of the cross-section width of the gasket **110**. The second longitudinal axis **114** is generally parallel to the first longitudinal axis **108**, and, as can be seen in FIG. **6**, is offset or spaced from the first longitudinal axis **108**. The distance of this offset can be at least 5 mm, and no greater than the cross-sectional width **126** (FIG. **8**) of the gaskets, which is about 30 mm, and will often be about 10-15 mm.

(39) As can be seen in FIG. **6**, the sealing portion **104** of the container gasket **102** and the sealing region **112** of the lid gasket **110** are compressed against each other to form the releasable seal between the lid **26** and the container **24**.

(40) Still in reference to FIG. **6**, the container gasket **102** includes a first rigid projection **120** along an outermost periphery of the container gasket **102**. The first rigid projection **120** can be generally parallel to the longitudinal axis **108**. Similarly, the lid gasket **110** includes a second rigid projection **122** along an innermost periphery of the lid gasket **110**. The second rigid projection **122** can be generally parallel to the longitudinal axis **114**. When the container gasket **102** and lid gasket **110** are compressed against each other, a first rigid projection **120** is pressed into the sealing region **112** of the lid gasket **110**, and the second rigid projection **122** is pressed into the sealing portion **104** of the container gasket **102**. This pressing of the respective rigid projections **120**, **122** into the opposing gasket member helps create a reliable, releasable, and repeatable seal.

(41) By reviewing FIGS. **3** and **6**, it should be appreciated that the gas inlet **56** and gas outlet **58** of the lid **26** are within the perimeter of the seal arrangement **100**.

(42) FIGS. **8-10** show an example embodiment for the container gasket **102** and lid gasket **110**. In this example, the container gasket **102** and the lid gasket **110** have the same geometry and dimensions. In other embodiments, there can be variations between the two.

(43) In this example, each of the container gasket **102** and lid gasket **110** has a thickness t (FIG. **9**) of about 10-30 mm, or about 20 mm; an overall length width w (FIG. **8**) of about 500-600 mm; and a length l of about 350-450 mm. The cross-sectional width at **126** can be about 10-30 mm, or about 20 mm. Each can be shaped to have a closed perimeter (closed outermost periphery) formed from two straight parallel segments, with two straight parallel extensions extending between the two segments, i.e. having a generally rectangular in shape but with the outside and inside corners

truncated, to form flat surfaces **124**. The angle **128** of the truncated corners to form the flat surfaces **124** can be about 40-50°, e.g. about 45°.

(44) Preferably, the container gasket **102** and lid gasket **110** are made from a single piece of material. For example, they can be cut from a material of closed cell rubber. Other ways of making the gaskets **102**, **110**, are possible, including extrusion or molding.

(45) In general, the gaskets **102**, **110** will be formed to maintain a functional compression with 3-6 mm compression. The gaskets can withstand a temperature of up to 120° C. the pressure range can be from -200 mBar up to +200 mBar. They will have the durability to allow up to 600 compressions.

(46) The above materials can be used in a method of sealing a filter housing, such as the filter housing **20**. The method includes covering the lid **26** over the container **24** to compress the sealing portion **104** of the container gasket **102** and the sealing region **112** of the lid gasket **110** against each other to form a releasable seal between the lid **26** and the container **24**. In doing so, the first longitudinal axis **108** and the second longitudinal axis **114** of the container gasket **102** and lid gasket **110** are offset from each other.

(47) The method can further include engaging the clamp arrangement **90** to provide a releasable clamping force between the container **24** and the lid **26**.

(48) The step of forming the seal can further include pressing the first rigid projection **120** of the container gasket **102** into the sealing region **112** of the lid gasket **110**, and pressing the second rigid projection **122** into the sealing portion **104** of the container gasket **102**.

(49) The method can further include moving the lid **26** by grasping the handle **60** secured to the lid **26**.

Inventive Aspects

(50) Inventive aspects of this disclosure include:

(51) Aspect 1: A filter housing comprising: (a) a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume; (b) a filter element in the interior volume; the filter element having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; (c) a lid removably mounted on the container to cover the access opening; and (d) a gas inlet and gas outlet in at least one of the container and lid; the gas inlet and gas outlet being co-planar; wherein the gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.

(52) Aspect 2. The filter housing of aspect 1 wherein the lid includes the gas inlet and the gas outlet.

(53) Aspect 3. The filter housing of any one of aspects 1 and 2 wherein: (a) the gas outlet includes an outlet tube extending into the interior volume of the container; and (b) the filter element has an outlet opening receiving the outlet tube.

(54) Aspect 4. The filter housing of aspect 3 wherein the filter element includes: (a) a media pack having the inlet face and the outlet face; and (b) a casing holding the media pack; wherein the casing includes an outlet wall spaced from the outlet face; the outlet wall defining the outlet opening of the filter element.

(55) Aspect 5. The filter housing of aspect 4 wherein the casing has a surrounding wall extending between the outlet wall and the inlet face of the media pack.

(56) Aspect 6. The filter housing of aspect 5 wherein the surrounding wall of the casing provides a baffle for inlet flow from the gas inlet to the inlet face of the media pack.

(57) Aspect 7. The filter housing of any one of aspects 4-6 wherein the media pack comprises pleated media.

(58) Aspect 8. The filter housing of any one of aspects 1-7 wherein the gas inlet and the filter element are arranged such that inlet gas turns between 160°-200° to flow from the gas inlet to the inlet face of the filter element.

- (59) Aspect 9. The filter housing of any one of aspects 1-8 wherein the container includes a ledge protruding into the interior volume constructed and arranged to hold the filter element.
- (60) Aspect 10. The filter housing of any one of aspects 1-9 wherein: (a) the container includes a filtering compartment and a debris collector disposed below the filtering compartment; (i) the filtering compartment and debris collector separated by a sloping wall having an aperture; the aperture covered by an openable and closeable fitment.
- (61) Aspect 11. The filter housing of any one of aspects 1-10 further including a seal arrangement between the compartment and the lid, and a clamp arrangement providing a releasable clamping force holding the compartment and lid together in sealing engagement.
- (62) Aspect 12. A method of filtering a gas comprising: (a) providing a container, a filter element in an interior volume of the container; and a lid mounted on the container; and (b) allowing gas to flow through a gas inlet, turn at least 90°, flow into the filter element, and then exit through a gas outlet; wherein the inlet and gas outlet are co-planar.
- (63) Aspect 13. The method of aspect 12 wherein the gas turns between 160°-200° to flow into the filter element.
- (64) Aspect 14. The method of any one of aspects 12 and 13 wherein the filter element includes a media pack having the inlet face and the outlet face; and a casing holding the media pack; wherein the casing has an outlet wall and a surrounding wall, the surrounding wall extending between the outlet wall and the inlet face of the media pack; and the step of allowing gas to flow includes the surrounding wall being a baffle for inlet flow from the gas inlet to the inlet face of the media pack.
- (65) Aspect 15. The method of any one of aspects 12-14 wherein the container includes a filtering compartment and a debris collector disposed below the filtering compartment; and while the gas flows from the gas inlet to the filter element, debris drops by gravity to a bottom of the filtering compartment.
- (66) The above represents example principles. Many embodiments can be made using these principles.

Claims

1. A filter housing comprising: (a) a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume; (b) a filter element in the interior volume; the filter element including a media pack having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; the filter element including a casing holding the media pack, wherein the casing includes an outlet wall spaced from the outlet face; the outlet wall defining an outlet opening of the filter element; (c) a lid removably mounted on the container to cover the access opening; and (d) a gas inlet and gas outlet in at least one of the container and lid; the gas inlet and gas outlet being co-planar; (i) the gas outlet includes an outlet tube extending into the interior volume of the container; and (ii) the filter element outlet opening receiving the outlet tube; wherein the gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.
2. The filter housing of claim 1 wherein the lid includes the gas inlet and the gas outlet.
3. The filter housing of claim 1 wherein the casing has a surrounding wall extending between the outlet wall and the inlet face of the media pack.
4. The filter housing of claim 3 wherein the surrounding wall of the casing provides a baffle for inlet flow from the gas inlet to the inlet face of the media pack.
5. The filter housing of claim 1 wherein the media pack comprises pleated media.
6. The filter housing of claim 1 wherein the gas inlet and the filter element are arranged such that inlet gas turns between 160°-200° to flow from the gas inlet to the inlet face of the filter element.
7. The filter housing of claim 1 wherein the container includes a ledge protruding into the interior

volume constructed and arranged to hold the filter element.

8. The filter housing of claim 1 wherein: (a) the container includes a filtering compartment and a debris collector disposed below the filtering compartment; (i) the filtering compartment and debris collector being separated by a sloping wall having an aperture; the aperture covered by an openable and closeable fitment.

9. The filter housing of claim 1 further including a seal arrangement between the container and the lid, and a clamp arrangement providing a releasable clamping force holding the container and lid together in sealing engagement.

10. A filter housing comprising: (a) a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume; (b) a filter element in the interior volume; the filter element including a media pack having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; the filter element including a casing holding the media pack, wherein the casing includes an outlet wall spaced from the outlet face; (c) a lid removably mounted on the container to cover the access opening; and (d) a gas inlet and gas outlet in the lid; the gas inlet and gas outlet being co-planar; the gas inlet having a first central axis in a direction of gas flow, and the gas outlet having a second central axis in the direction of gas flow; the first central axis and second central axis being parallel to each other; wherein the gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.

11. The filter housing of claim 10 wherein the gas inlet and the filter element are arranged such that inlet gas turns between 160° - 200° to flow from the gas inlet to the inlet face of the filter element.

12. The filter housing of claim 10 wherein the container includes a ledge protruding into the interior volume constructed and arranged to hold the filter element.

13. The filter housing of claim 10 wherein: (a) the container includes a filtering compartment and a debris collector disposed below the filtering compartment; (i) the filtering compartment and debris collector being separated by a sloping wall having an aperture; the aperture covered by an openable and closeable fitment.

14. The filter housing of claim 10 further including a seal arrangement between the container and the lid, and a clamp arrangement providing a releasable clamping force holding the container and lid together in sealing engagement.

15. A filter housing comprising: (a) a container with a surrounding wall defining an open interior volume; the container having an access opening in communication with the interior volume; (i) the container includes a filtering compartment and a debris collector disposed below the filtering compartment; the filtering compartment and debris collector being separated by a sloping wall having an aperture; the aperture covered by an openable and closeable fitment; (b) a filter element in the interior volume; the filter element having an inlet face and outlet face and oriented for filtering gas by gas flow through the inlet face and out through the outlet face; (c) a lid removably mounted on the container to cover the access opening; and (d) a gas inlet and gas outlet in at least one of the container and lid; the gas inlet and gas outlet being co-planar; (i) the gas outlet includes an outlet tube extending into the interior volume of the container; and (ii) the filter element outlet opening receiving the outlet tube; wherein the gas inlet and the filter element are arranged such that inlet gas turns at least 90° to flow from the gas inlet to the inlet face of the filter element.

16. The filter housing of claim 15, wherein the lid includes the gas inlet and the gas outlet.

17. The filter housing of claim 15 wherein the container includes a ledge protruding into the interior volume constructed and arranged to hold the filter element.

18. The filter housing of claim 15 further including a seal arrangement between the container and the lid, and a clamp arrangement providing a releasable clamping force holding the container and lid together in sealing engagement.
